

OUTREACH NODE — CUSTOM COLD EMAILER

Multi-Agent AI Outreach Orchestrator & Voicemail Synthesizer

UAV Lab Internship Capstone Project • Developer: Mughees Tayyab • Status: Production Release

Executive Summary & Project Overview

Outreach Node (GreenFactor) is a state-of-the-art, multi-agent cold email campaign manager engineered to automate personalized outbound prospecting. The project solves the key limitations of traditional cold emailing—namely, static templates, manual company research, and high bounce rates—by orchestrating a structured sequence of AI agents. Each agent operates with specific roles, performing keyless scraping, company profiling, copywriting, and recursive critique loops. Additionally, the system features Google Text-to-Speech (gTTS) integration to automatically synthesize customized audio voicemail pitches, and an SMTP email dispatcher to support one-click campaign execution. The interface utilizes a premium dark glassmorphism design system to visualize execution pipelines and results.

Key System Capabilities

- **Social Media Prospecting:** DuckDuckGo-powered keyless lookups discover professional summaries and organizational titles dynamically without relying on expensive, rate-limited LinkedIn APIs.
- **Company Context Ingestion:** Synthesizes strategist briefs by scraping Wikipedia and executing targeted search queries to extract recent news, core offerings, and strategic pain points.
- **Personalized Copywriting:** Generates custom subject lines and body copy tailored to specific target alignment hooks, brand tones, and sender profiles.
- **Self-Correcting Critique Loop:** The proofreader agent evaluates drafts against strict alignment and accuracy criteria. If a draft scores below 7/10, the proofreader generates feedback, prompting the copywriter to revise up to 3 times.
- **Voicemail Synthesis (gTTS):** Automatically generates 100% free personalized MP3 voice scripts referencing the prospect's company and sender context, eliminating the need for paid speech synthesis API keys.
- **SMTP Dispatcher & Excel Reporting:** Supports direct campaign dispatching via SMTP (e.g., Gmail App Passwords) and compiles a fully formatted, color-coded Excel spreadsheet output mapping campaign progress.

Technical Infrastructure & Requirements

Category	Dependencies / Technologies	System Role
Core Framework	Python 3.10+, Flask 3.1.1, python-dotenv	Handles backend server operations, configuration, and routing.
AI Client Layer	litellm, google-genai, openai	Provides a unified API client wrapping model invocation (gateway-claude-opus-4-8).
Intelligence Tools	duckduckgo-search, requests, lxml	Executes search scraping and extracts text from Wikipedia pages.
Data Processing	openpyxl, pandas, PyPDF2, python-docx	Parses incoming context PDFs/DOCX and outputs styled Excel logs.
Media & Dispatch	gTTS 2.5.4, smtplib (SMTP)	Renders personalized MP3 voicemail audio and dispatches emails.

System Architecture & Multi-Agent Pipeline

The core logic of Outreach Node is encapsulated in the Orchestrator class, which runs each campaign asynchronously in dedicated background threads to prevent UI-blocking. A centralized state management engine (StateManager) records execution logs as JSON records and handles concurrent pipeline progress updates in real-time.

Sequential Execution Sequence

- Stage 0: Orchestrator Agent (Analysis):** Before querying individual prospects, the Orchestrator Agent analyzes the user's custom campaign prompt and attached reference documents. It generates a structured `research_plan` defining target keywords, LinkedIn/Web search query templates, and copy requirements.
- Stage 1: Prospecting Agent (Discovery):** Extracts the target name and company from the input sheet, running keyless web queries to output a high-level summary of the prospect's professional background.
- Stage 2: LinkedIn Agent (Intelligence):** Guided by the `research_plan`, it performs targeted searches to discover additional professional insights, target milestones, and executive context.
- Stage 3: Context Agent (Company Profiling):** Gathers deep corporate details by combining Wikipedia scrapes with DuckDuckGo searches, mapping the company's core services, recent news, and specific pain points.
- Stage 4: Copywriter Agent (Draft Generation):** Synthesizes all gathered information (prospect summary, corporate pain points, sender credentials, tone settings, and campaign goals) to draft a hyper-personalized email subject and body.
- Stage 5: Proofreader Agent (Critique & Self-Correction):** Analyzes the draft against ground truths and alignment guidelines, scoring the copy on Relevance, Tone, Personalization, and Accuracy. If the score is < 7 , it generates specific critiques and returns execution to the Copywriter for revision (capped at 3 retries).
- Stage 6: Output Compilation (Excel & TTS):** On final draft approval, the system updates the campaign log, runs gTTS to generate the MP3 voicemail file, and writes all records to a formatted Excel file.

Software Architecture & Directory Blueprint

Directory / File	Code Files	Operational Implementation & Design
agents/	orchestrator_agent.py, context_agent.py, copywriter_agent.py, proofreader_agent.py	LLM agents implementing tailored prompts, JSON output formatting, and recursive validation loops.
middleware/	orchestrator.py, state_manager.py, ai_client.py	Campaign orchestration thread managers, JSON state managers, and central LLM connection middleware.
tools/	search_tool.py, excel_tool.py, tts_tool.py, document_tool.py, email_tool.py	Underlying functional APIs for web scraping, Excel openpyxl parsing, TTS audio rendering, and SMTP emailing.
templates/ & static/	dashboard.html, pipeline.html, results.html, index.css	Dark glassmorphism frontend layout, stepper trackers, audio players, and inline copy editors.

Verification & Deployment

- 1. Setup & Launch:** Initialize python environment: `python -m venv venv`, activate it, and run `pip install -r requirements.txt`. Copy `.env.example` to `.env` and populate `API_KEY` and optional SMTP settings. Launch via `python app.py`.
- 2. Automated Tests:** Run test suites locally using: `pytest tests/ -v`. Tests validate the Mock SMTP dispatcher, state logging serialization, Excel parsers, and individual agent logic.